# Assignment 3 Report

Course 203.3630 — Artificial Intelligence Lab.

All numbers in this report come from the generated result files under `results/final_experiments/` (final run with seeds 42, 43, 44 from `configs/final_experiment_plan.json`). Nothing was filled in by hand without a matching CSV row; `scripts/extract_report_numbers.py` prints the same facts for checking.

## 1. Introduction

The assignment has two parts. Part A asks for six search algorithms (Simulated Annealing, Tabu Search, Ant Colony Optimization, a Genetic Algorithm with an Island Model, Adaptive Large Neighborhood Search, and Branch & Bound / Limited Discrepancy Search), first validated on the continuous Ackley function as a warm-up and then compared on six official CVRP benchmark instances, together with an explicit multi-stage heuristic. Part B asks for generative AI: evolving Rush Hour heuristic functions with both Genetic Programming (GP) and Gene Expression Programming (GEP), where every candidate heuristic is judged by actually running A* with it.

All stochastic runs use the fixed seeds 42, 43 and 44 from the final experiment plan, so every number below is reproducible.

## 2. Implementation Overview

### 2.1 Project structure

The code is a plain Python package: `src/common/` (timing), `src/ackley/` (function + the six adaptations), `src/cvrp/` (model, parser, cost, validation, baseline) with `src/cvrp/solvers/` (the six algorithms), `src/rushhour/` (board, A*, safe evaluator, GP/GEP comparison), `src/gp/` and `src/gep/` (separate frameworks), and `src/experiments/` (runners, summaries, report assets). `scripts/` holds the CLI entry points and `tests/` the pytest suite (268 tests at the time of the final run).

### 2.2 Reproducibility and command-line interface

Every run takes its inputs from CLI arguments — instance paths, seeds, budgets and timeouts are never hardcoded. Each experiment row is written to CSV with its seed, budget, timeout, costs, feasibility, errors and timing.

The final settings live in `configs/final_experiment_plan.json` and the runner `scripts/run_final_experiments.py` is resumable (finished raw CSVs are skipped on a rerun).

### 2.3 Validation and safety checks

Every CVRP solution is validated: routes start and end at the depot, no customer is missing or duplicated, capacity is respected, and the number of used routes must not exceed the vehicle count. Rows report `feasible` and the exact validation errors — nothing is silently fixed. Rush Hour heuristic evaluation runs A* under a node cap, a per-puzzle time cap and a total time budget, and an exception inside a candidate heuristic is recorded as a failed puzzle instead of crashing the run.

## 3. Part A — Ackley Function

The Ackley function for d dimensions:

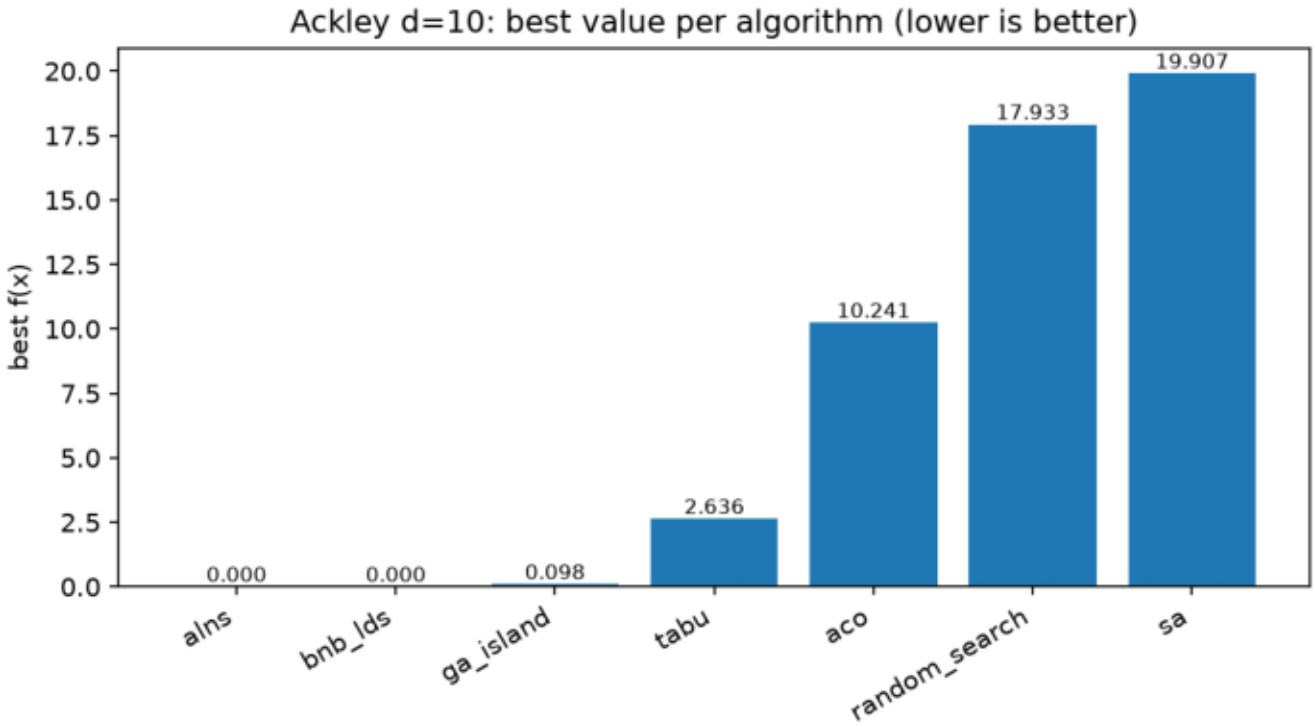
```
f(x) = -a * exp(-b * sqrt((1/d) * sum(x_i^2)))
       - exp((1/d) * sum(cos(c * x_i))) + a + e
```

with a = 20, b = 0.2, c = 2*pi, dimension d = 10, bounds
x_i in [-32.768, 32.768], and the known optimum f(0, ..., 0) = 0.

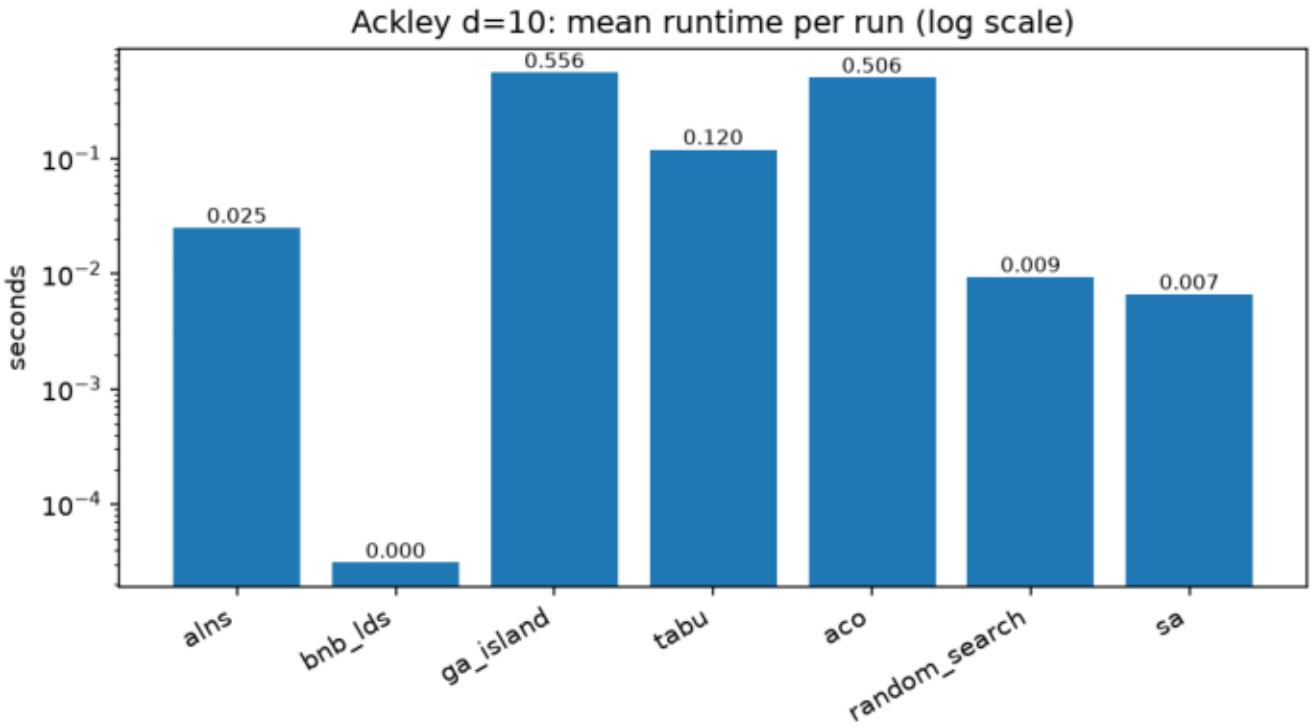
Final results (3 seeds per algorithm, budget 500, timeout 60 s), from

`summary/ackley_d10_summary.csv`:

| algorithm | runs | best_value | mean_best_value | std_best_value | mean_dist_from_origin | mean_elapsed (s) |
|---|---|---|---|---|---|---|
| ackley_alns | 3 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.025 |
| ackley_bnb_lds | 3 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000 |
| ackley_ga_island | 3 | 0.098 | 0.100 | 0.002 | 0.063 | 0.556 |
| ackley_tabu | 3 | 2.636 | 3.018 | 0.359 | 1.466 | 0.120 |
| ackley_aco | 3 | 10.241 | 11.763 | 1.406 | 11.187 | 0.506 |
| random_search | 3 | 17.933 | 18.320 | 0.481 | 30.278 | 0.009 |
| ackley_sa | 3 | 19.907 | 20.041 | 0.118 | 46.706 | 0.007 |




The figure shows the best value each algorithm reached over its three seeds,
sorted from best to worst. The spread is huge — from exactly 0 to almost 20 —
which mostly reflects how well each adaptation fits a continuous function,
not a universal ranking of the methods.




Runtime (log scale) shows the other side: SA and random search are almost free, while ACO and GA-Island pay for population/colony bookkeeping. Note that B&B/LDS looks instant only because its greedy zero-first bin order finds the origin immediately and the search stops early.

Short analysis. ALNS and B&B/LDS reached 0.000000 (six decimals) on every seed — but both benefit from knowing the optimum is at the origin: the ALNS "toward zero" repair operator and the B&B/LDS bin ranking by distance to zero point straight at it, so this says more about the adaptation than about general optimization strength. GA-Island got close (?0.1) without such a hint. Tabu Search reached ?2.6. ACO's coarse bins (10 per dimension) limit its precision. SA with the untuned parameters (initial temperature 10, step scale 1) actually ended worse than plain random search — an honest negative result: 500 local Gaussian steps from a random corner of a 10-dimensional box are simply not enough without parameter tuning.

## 4. Part A — CVRP

Official instances and best-known solutions (BKS):

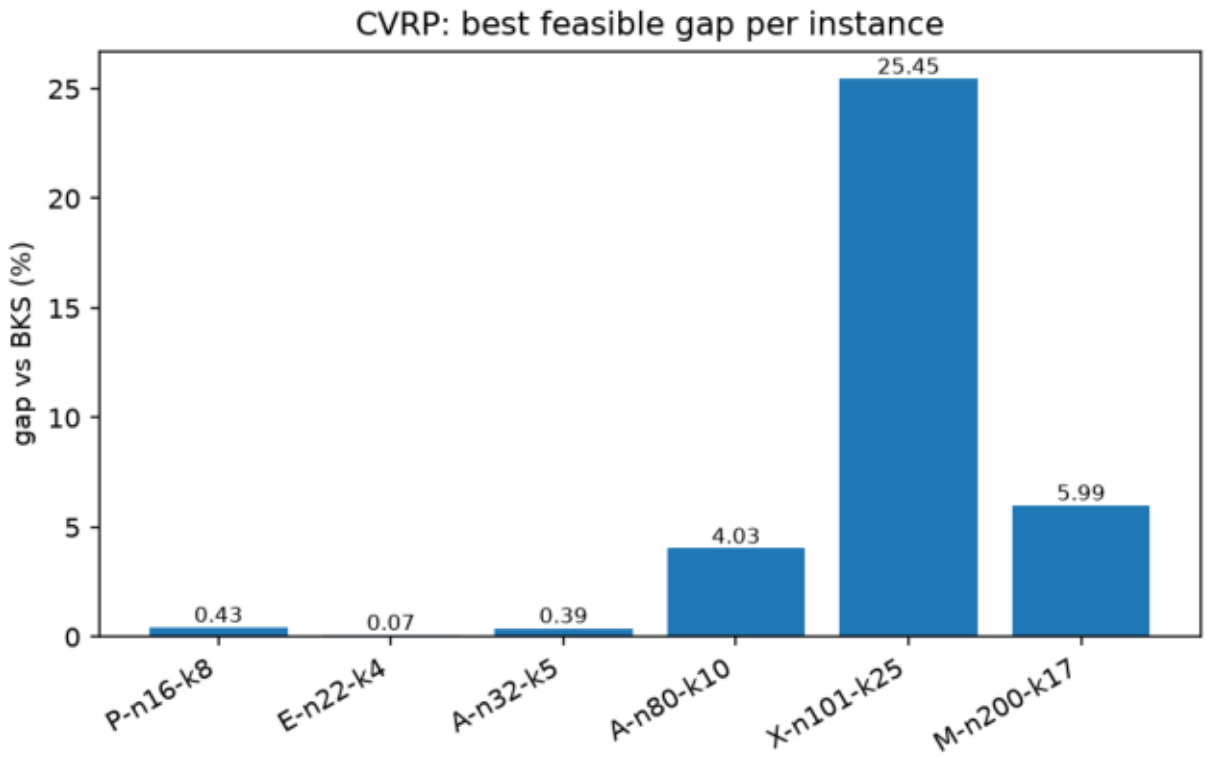
| instance | BKS cost |
| --- | --- |
| P-n16-k8 | 450 |
| E-n22-k4 | 375 |
| A-n32-k5 | 784 |
| A-n80-k10 | 1763 |
| X-n101-k25 | 27591 |
| M-n200-k17 | 1275 |

The BKS values are used only to compute the gap percentage (100 · (cost ? BKS) / BKS); the algorithms never see them.

Best feasible result per instance (7 algorithms × 3 seeds = 21 rows per instance; after the Stage 9-B2 repair **all 126 rows are feasible**):

| instance | BKS | best algorithm | best cost | best gap | feasible runs |
| --- | --- | --- | --- | --- | --- |
| P-n16-k8 | 450 | tabu | 451.95 | 0.43% | 21/21 |
| E-n22-k4 | 375 | alns | 375.28 | 0.07% | 21/21 |
| A-n32-k5 | 784 | alns | 787.08 | 0.39% | 21/21 |
| A-n80-k10 | 1763 | alns | 1834.01 | 4.03% | 21/21 |

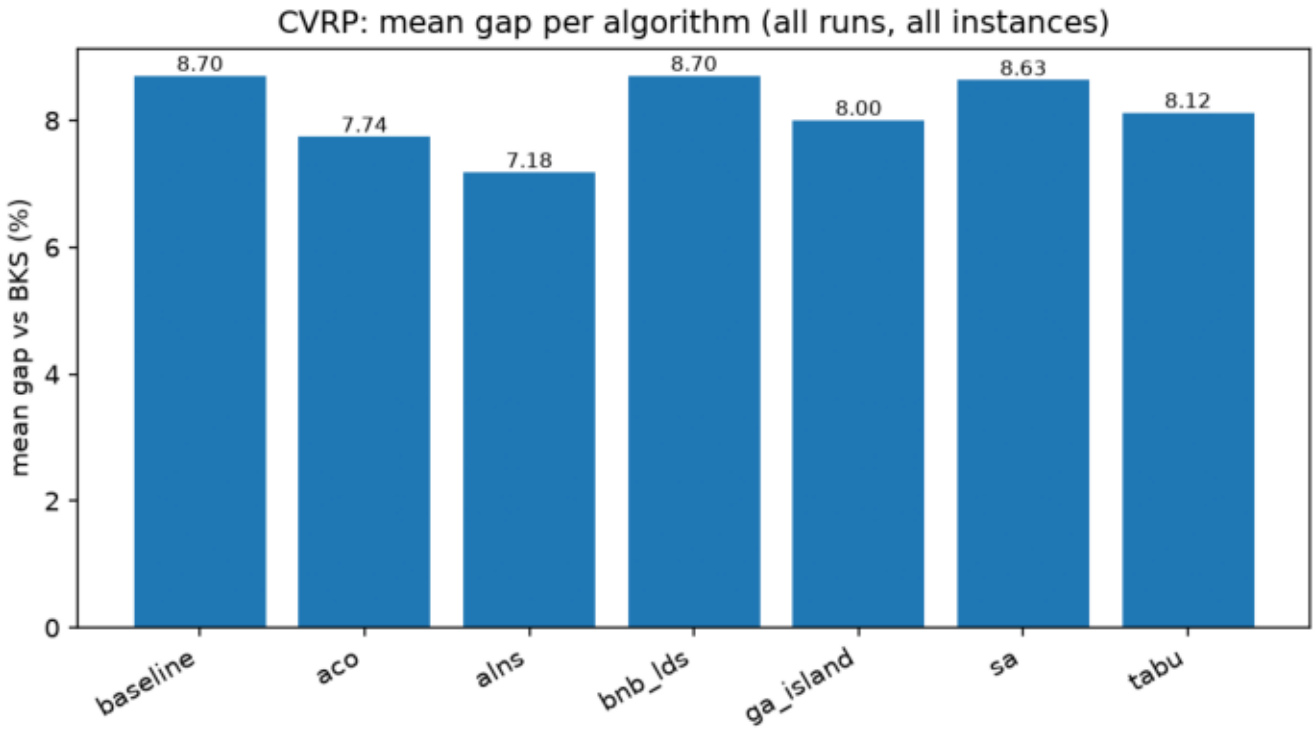
| X-n101-k25 | 27591 | aco | 34613.14 | 25.45% | 21/21 |
| M-n200-k17 | 1275 | alns | 1351.43 | 5.99% | 21/21 |



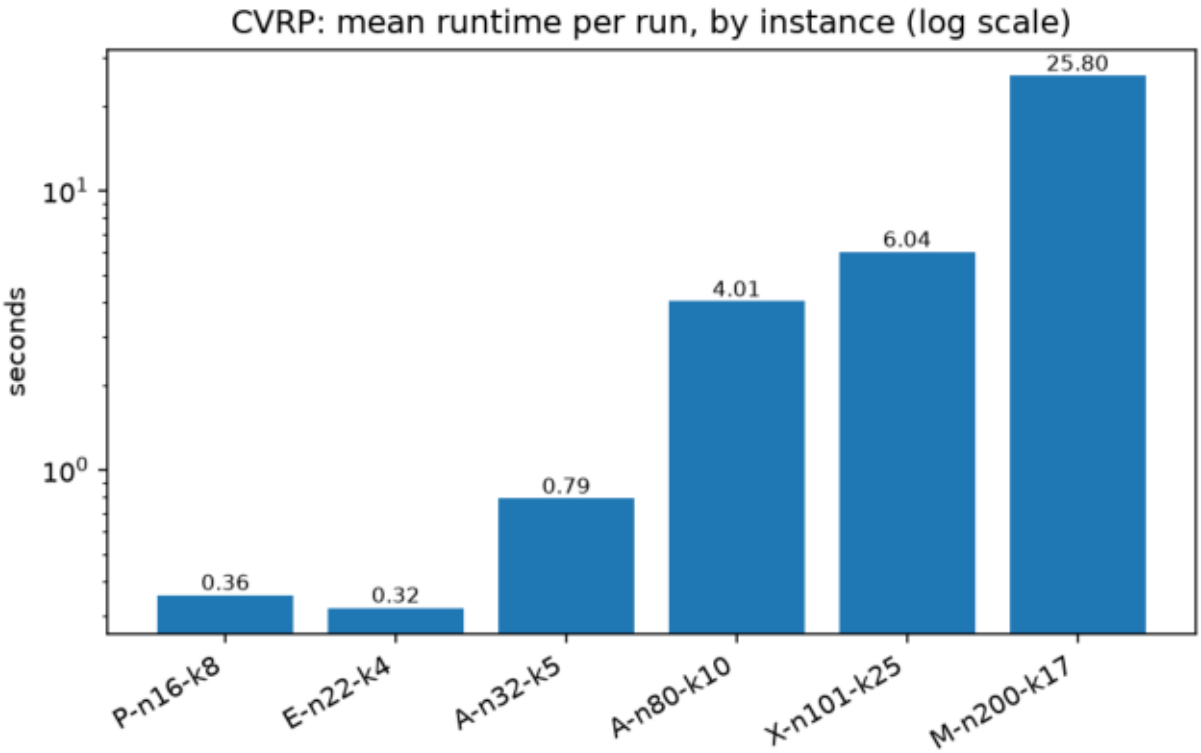

The picture is very clear: the three small instances end below half a percent from the best known solutions, the two larger ones land at 4–6%, and X-n101-k25 stands out at 25.45%. That column is not an algorithm bug — it is a capacity-packing property of the instance, explained below. It should not be read as "the algorithms failed"; it should be read as "this instance leaves no room for the local moves these algorithms rely on".

Per algorithm (mean gap over all 18 runs each, plus where it was strongest and weakest):

| algorithm | mean gap (all runs) | best instance | weakest instance |
| --- | --- | --- | --- |
| alns | 7.18% | E-n22-k4 (0.07%) | X-n101-k25 (27.08%) |
| aco | 7.74% | P-n16-k8 (0.43%) | X-n101-k25 (25.45%) |
| ga_island | 8.00% | P-n16-k8 (0.43%) | X-n101-k25 (25.74%) |
| tabu | 8.12% | P-n16-k8 (0.43%) | X-n101-k25 (26.82%) |
| sa | 8.63% | P-n16-k8 (1.55%) | X-n101-k25 (26.95%) |
| baseline | 8.70% | P-n16-k8 (2.65%) | X-n101-k25 (27.08%) |
| bnb_lds | 8.70% | P-n16-k8 (2.65%) | X-n101-k25 (27.08%) |




Averaged over all runs and instances, the ordering is ALNS < ACO < GA-Island < Tabu < SA < baseline = B&B/LDS. The differences are small (7.2%–8.7%) because the X instance dominates every mean; on the other five instances the ordering is similar but the absolute gaps are much smaller. B&B/LDS matching the baseline exactly is an honest result: within its node and discrepancy limits it never found anything better than its starting incumbent.




Mean runtime per run grows with instance size as expected (log scale). The

values are averages over all seven algorithms, so the cheap methods (SA, baseline) pull the mean down; ACO alone accounts for most of the time on the two largest instances.

Honest note on X-n101-k25: its total demand is 5147 while the total fleet capacity is 25 × 206 = 5150 — only 3 units of slack over 25 routes, so almost every route must be loaded completely full. Clarke-Wright needed 28 routes there, and a subset-sum packing repair (Section 5) was required just to reach feasibility. That packing ignores geometry, and with nearly zero capacity slack the usual local moves (relocate, cross-route swap) are almost always capacity-infeasible, so the metaheuristics could barely improve the start. The 25.45% gap is real and reported as such.

## 5. Multi-stage CVRP Heuristic

The baseline heuristic runs in five stages:

1. **Construction** with Clarke-Wright savings (merge route ends by descending saving under the capacity limit).
2. **2-opt** inside each route (reverse inner segments while improving).
3. **Relocate** between routes (move single customers while improving).
4. **Vehicle-count repair**: first empty surplus routes and reinsert their customers at cheapest feasible positions; if that fails, deterministic rebuilds (cheapest insertion, best-fit packing, sweep by polar angle); as a last resort a subset-sum packing that fills vehicles one at a time as full as possible over the integer demands — this is what makes X-n101-k25 feasible.
5. **Final validation** — a failed repair is returned as feasible=False with errors, never hidden.

Why the repair stage was necessary: Clarke-Wright merges routes only while the merge is profitable, so on instances with a tight fleet it can stop with more routes than vehicles exist. This actually happened twice in the final runs — P-n16-k8 (9 routes for 8 vehicles) and X-n101-k25 (28 routes for 25). Without the repair, every algorithm that starts from the baseline would inherit an infeasible incumbent, which is exactly what the first final run showed before stages 8-B2/9-B2. The subset-sum packing below is the piece that finally made X-n101-k25 feasible:

src/cvrp/local_search.py — subset-sum vehicle repair

```
349 | def build_routes_subset_sum_packing(instance, customer_order):
350 |     """Fill vehicles one at a time as full as possible, using a subset-sum
351 |     table over the integer demands.
352 |
353 |     Needed when total demand almost equals total capacity: X-n101-k25 has
354 |     5147 demand against 25 * 206 = 5150 capacity, so almost every route must
355 |     be loaded completely full, which greedy packing cannot find.
356 |     Returns customer groups or None.
357 |     """
358 |     capacity = int(instance.capacity)
359 |     if instance.capacity != capacity:
360 |         return None  # table needs integer capacity
361 |     demands = {}
362 |     for customer in customer_order:
363 |         demand = instance.demands.get(customer, 0.0)
364 |         if demand != int(demand) or demand > capacity:
365 |             return None  # table needs integer demands
366 |         demands[customer] = int(demand)
367 |
368 |     remaining = list(customer_order)
369 |     total_left = sum(demands[c] for c in remaining)
370 |     groups = []
371 |     for bins_left in range(instance.vehicle_count, 0, -1):
372 |         if not remaining:
373 |             break
374 |         # everything not in this bin must still fit into the other bins
375 |         lower = max(0, total_left - (bins_left - 1) * capacity)
376 |         if lower > capacity:
```

The idea is simple to state: when total demand almost equals total fleet capacity, almost every vehicle must leave completely full, so the repair fills vehicles one at a time with a subset of customers whose demands sum as close to the capacity as possible (a small dynamic-programming table over the integer demands), with a lower bound making sure the remaining customers still fit into the remaining vehicles.

Complexity: everything here is heuristic, not exact. One 2-opt pass over a route of length L costs $O(L^2)$ route evaluations and passes repeat until no improvement. The relocate pass scans all customer/position pairs, roughly $O(n^2)$ per pass. The repair adds packing work (the subset-sum table is $O(n \cdot \text{capacity})$ per vehicle) but guarantees the route count fits the fleet, which turned out to be essential on two of the six official instances.

## 6. CVRP Algorithms

All six start from the multi-stage baseline solution, share one random relocate/swap/2-opt neighborhood where applicable, and were run with the same per-instance budget and timeout for a fair comparison.

### 6.1 Simulated Annealing

Full solutions as states; one random neighbor per iteration; accept improvements always and worse candidates with probability exp(-delta/T), T multiplied by 0.995 per iteration. SA improved the small instances (best P-n16-k8 gap 1.55%) but with this untuned schedule it mostly stayed near the baseline on the larger ones (mean gap 8.63%).

### 6.2 Tabu Search

Samples 40 candidates per iteration from the shared neighborhood, forbids

recently visited solutions via a customer-sequence signature (FIFO tenure 30), with aspiration on a new global best. Tied for the best P-n16-k8 result (0.43%) and was solid on the small instances; mean gap 8.12%.

### 6.3 Ant Colony Optimization

Ants build routes customer by customer with probability proportional to pheromone^? · (1/distance)^? under the capacity limit, light 2-opt per ant, evaporation plus deposits on the iteration and global best. ACO produced the best feasible X-n101-k25 result (25.45%) — its constructive nature sidesteps the frozen-local-moves problem there — at the price of the highest runtime (mean 32.8 s per run).

### 6.4 GA Island Model

Giant-tour chromosomes (customer permutations) split into routes by a capacity-aware scan; OX crossover, swap/inversion mutation, tournament selection, elitism 1; two islands with ring migration every 20 generations. Solid mid-field (mean gap 8.00%), best on P-n16-k8 (0.43%).

### 6.5 ALNS

Destroy operators (random and worst removal) and repair operators (greedy and regret-2 insertion) chosen by adaptive weights, with simulated-annealing acceptance. ALNS was the strongest method overall in these runs: best feasible result on four of six instances (E-n22-k4 0.07%, A-n32-k5 0.39%, A-n80-k10 4.03%, M-n200-k17 5.99%) and the lowest mean gap (7.18%) at a moderate mean runtime of 5.2 s.

### 6.6 Branch-and-Bound / LDS

A time-limited search over customer insertion decisions (hardest customers first), where taking the k-th cheapest insertion costs k discrepancy, with a partial-cost bound against the incumbent. It is exact-inspired, not a full exact solver for these sizes: within the node/discrepancy limits it never beat the baseline incumbent on any official instance, so its rows match the baseline (mean gap 8.70%). It proved useful mainly as a correctness reference.

## 7. Ackley Adaptations

SA and Tabu Search are natural continuous methods here (Gaussian steps; a tabu list of rounded points). GA-Island and ALNS operate directly on continuous vectors (blend crossover and Gaussian mutation; dimension-wise destroy and repair). ACO and B&B/LDS are honest discretizations (pheromone over bins per dimension; limited-discrepancy search over bin choices) — the canonical versions of these methods are combinatorial, and the report does not claim otherwise. Random search is only a sanity baseline. The results in Section 3 reflect exactly this: the methods whose adaptation points toward the origin (ALNS's toward-zero repair, B&B/LDS's zero-first bin order) reached it, the general-purpose ones (GA-Island, Tabu) got close, and untuned SA performed worst.

## 8. Part B — Rush Hour with GP and GEP

A* solves 6×6 Rush Hour boards with g = number of moves and a pluggable heuristic h. GP and GEP evolve h as expressions over three board features — distance of the red car to the exit, number of blocking cars, number of free exit cells — plus small constants, with protected operators (+, -, *, /, min, max, abs, neg, log). Fitness strongly rewards solved puzzles (10000 each) and penalizes expanded nodes, solution cost, timeouts and node-cap hits. Every evaluation runs under the safety caps from Section 2.3.

GP uses expression trees with subtree crossover and mutation. GEP is a

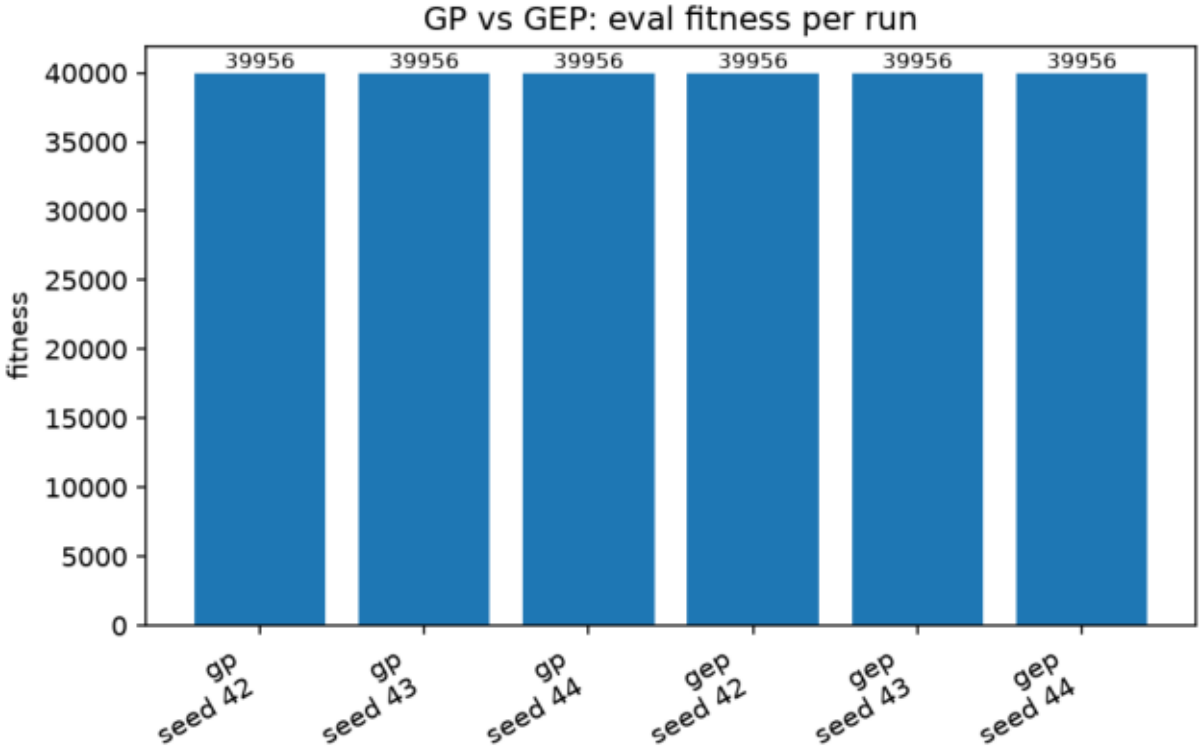
separate framework: a linear genome with a head (functions or terminals) and a tail (terminals only), decoded as a Karva K-expression; its operators are point mutation and one-/two-point crossover on the flat gene string. The decoder is the heart of GEP — it reads the linear genome left to right and attaches children level by level, ignoring leftover symbols:

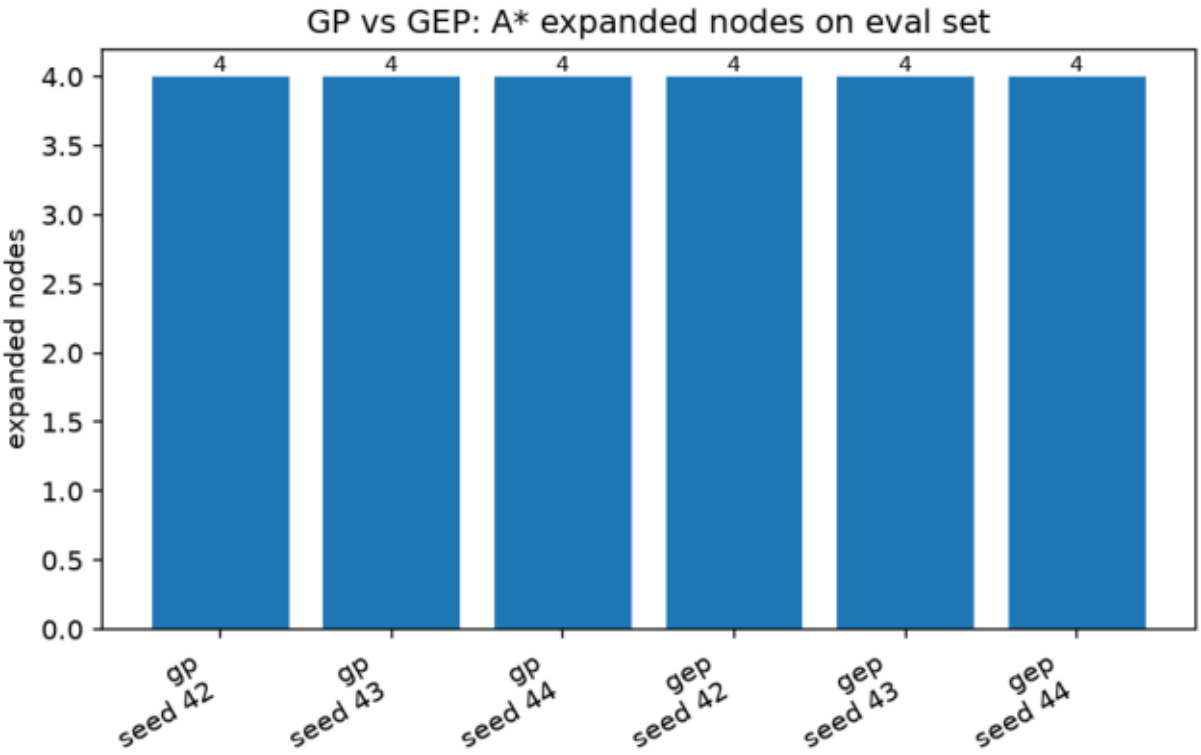
src/gep/decoder.py — Karva decoding

```
  27 | def decode_genome_to_tree(genome: GEPGenome) -> GPTree:
  28 |     symbols = genome.genes
  29 |     root = _make_node(symbols[0])
  30 |     open_nodes = deque()
  31 |     if ARITY.get(symbols[0], 0) > 0:
  32 |         open_nodes.append((root, ARITY[symbols[0]]))
  33 |
  34 |     index = 1
  35 |     while open_nodes:
  36 |         node, needed = open_nodes[0]
  37 |         if len(node.children) == needed:
  38 |             open_nodes.popleft()
  39 |             continue
  40 |         if index < len(symbols):
  41 |             symbol = symbols[index]
  42 |             index += 1
  43 |         else:
  44 |             symbol = "0.0"  # genome ended early, pad with a safe terminal
  45 |         child = _make_node(symbol)
  46 |         node.children.append(child)
  47 |         arity = ARITY.get(symbol, 0)
  48 |         if arity > 0:
  49 |             open_nodes.append((child, arity))
  50 |
  51 |     return GPTree(root=root)
  52 |
  53 |
```

Final comparison (train and eval sets of 4 puzzles each, 20 generations, population 30, seeds 42–44), from `raw/gp_gep_comparison_runs.csv`:

| algorithm | seed | eval fitness | solved | expanded | cost | best expression |
|---|---|---|---|---|---|---|
| gp | 42 | 39956 | 4/4 | 4 | 4 | min((min(blocking, blocking) * (distance - 0)), ((distance * blocking) / (blocking / distance))) |
| gp | 43 | 39956 | 4/4 | 4 | 4 | (blocking - min((2 / neg((1 + free))), blocking)) |
| gp | 44 | 39956 | 4/4 | 4 | 4 | max((log(distance) / neg(1)), max(min(5, blocking), (blocking * distance))) |
| gep | 42 | 39956 | 4/4 | 4 | 4 | (5 * blocking) |
| gep | 43 | 39956 | 4/4 | 4 | 4 | ((((1 + distance) * blocking) + 0.5) + 1) |
| gep | 44 | 39956 | 4/4 | 4 | 4 | (((blocking * 1) / (blocking - 0.5)) + (abs(2) - free)) |




The two charts make the "tie" visible: identical fitness (39956) and identical A* effort (4 expanded nodes total) in every run of both methods. What this comparison actually means: on this evaluation set the evolved heuristics guide A* essentially perfectly, so quality cannot separate GP from GEP here — the interesting differences are the shape of the evolved expressions and the evolution time. What it does not mean: that GP and GEP are equal in general. Four small puzzles are simply not enough pressure; a harder puzzle set would likely start separating the methods.

Summary. Every run (both methods, all seeds) solved 4/4 evaluation puzzles with only 4 expanded nodes total, giving the identical best fitness 39956 — the evaluation set is too easy to separate the methods on quality. Expression diversity is 1.00 for both (all best expressions differ across seeds), and GEP's genome diversity is also 1.00. Evolution time slightly favored GEP (?0.7–0.9 s per run vs ?1.1–1.4 s for GP), and GEP's best expressions are visibly shorter (e.g. `(5 * blocking)`). With only 4 tiny puzzles these are observations, not general conclusions.

## 9. Results

The main tables and figures are in Sections 3 (Ackley), 4 (CVRP) and 8 (GP/GEP); the committed report figures live under `report/figures/`. The raw per-run rows live under `results/final_experiments/raw/` (126 CVRP rows, 21 Ackley rows, 6 GP/GEP rows), the aggregated tables under `results/final_experiments/summary/`, and additional generated assets under `results/final_experiments/report_assets/`. The execution manifest (`final_execution_manifest.json`) records what ran, with which budgets, and how long it took.

## 10. Analysis and Discussion

- **CVRP pattern.** Gaps grow with instance size: under 0.5% on the three small instances, ~4–6% on A-n80-k10 and M-n200-k17. ALNS was the most consistent method; ACO was the most expensive but handled the tight X instance best.
- **X-n101-k25.** With 3 units of total capacity slack, feasibility is a bin-packing problem and quality improvement is nearly frozen: any relocate overfills a route. Repairing to feasibility cost geometry (packing ignores coordinates), leaving a 25.45% gap. Improving this would need capacity-aware compound moves (e.g. ejection chains), which were out of scope.
- **Ackley.** The warm-up separated the methods clearly, but part of that separation comes from how each was adapted (Section 7). Untuned SA losing to random search is a useful reminder that parameter choices matter as much as the algorithm name.
- **GP vs GEP.** Equal quality on this small benchmark; GEP was somewhat faster and produced more compact expressions; GP trees were larger. Both frameworks are genuinely different representations, not renames.
- **Runtime vs quality.** ACO bought its X result with ~33 s mean runtime; ALNS delivered the best average quality at ~5 s; SA is nearly free but weakest. B&B/LDS spent its budget without beating the incumbent.
- **Limitations.** One fixed budget/timeout profile per instance, three seeds, no tuning, and a small Rush Hour puzzle set — the comparisons hold for this setup only.

## 11. Complexity and Practical Considerations

- **Feasibility checking** is O(n) per solution and is run on every candidate the metaheuristics accept, plus once on every final result — the cost is small compared to neighborhood evaluation and it caught real bugs (route-count violations) that pure cost comparison would hide.
- **Neighborhood costs.** The shared relocate/swap/2-opt neighborhood costs O(n) per sampled neighbor (copy + cost); Tabu's 40 candidates per iteration make it ~40× SA per iteration, which matches the observed runtimes.
- **Stochastic variance.** Three seeds per configuration; the summary CSVs report mean and standard deviation. The seeds are fixed in the plan, so every row can be regenerated exactly.
- **Timeouts and fairness.** All algorithms get the same timeout on the same instance, so slower-per-iteration methods are not silently given more work. Budgets differ per algorithm in meaning (iterations vs generations vs nodes), which is why the timeout is the binding fairness control on the large instances.
- **No optimality claims.** Everything here is heuristic or time-limited;

the gaps against BKS quantify exactly how far the results are from the best known solutions.

## 12. Use of AI Tools

AI tools were used as coding and debugging assistants during the project. The final code and this report were reviewed and understood by the student.

Any generated code was tested and adjusted through the project's test suite and staged workflow.

## 13. Reproducibility

- Python 3.12.3, dependencies via `pip install -r requirements.txt` (numpy, matplotlib, pandas, pytest).
- Place the six official CVRPLIB files under `data/official_cvrp/` and verify with `python scripts/check_official_cvrp_data.py --strict`.
- Smoke check: `python scripts/run_smoke_suite.py --output-dir results/smoke_suite`
- Print/validate the final plan:

  `python scripts/print_final_experiment_plan.py --require-official-data`
- Full final run (resumable): `python scripts/run_final_experiments.py`
- Report figures: `python scripts/generate_report_figures.py`, then `python scripts/export_report_pdf.py` for this PDF
- Report facts: `python scripts/extract_report_numbers.py`
- All generated results stay under `results/final_experiments/` and are not committed to Git.

The final runner is what makes the experiments reproducible in practice: it validates the plan and the official data before anything runs, executes each part through the same Python functions the tests use, writes one CSV row per run with its seed/budget/timeout, and skips already-finished raw files — when the vehicle-count repair changed, only X-n101-k25 had to be rerun while the other five instances were reused untouched:

src/experiments/final_execution.py — final suite

```
260 | def run_final_experiment_suite(plan_path="configs/final_experiment_plan.json",
261 |                                resume: bool = True, run_cvrp: bool = True,
262 |                                run_ackley: bool = True,
263 |                                run_rushhour: bool = True) -> dict:
264 |     plan = load_final_plan(plan_path)
265 |     ok, errors = validate_final_plan(plan, require_official_data=True)
266 |     if not ok:
267 |         raise ValueError("final plan validation failed: " + " | ".join(errors))
268 |
269 |     output_dir = plan.get("output_dir", "results/final_experiments")
270 |     manifest = {
271 |         "plan_path": str(plan_path),
272 |         "resume": resume,
273 |         "run_cvrp": run_cvrp,
274 |         "run_ackley": run_ackley,
275 |         "run_rushhour": run_rushhour,
276 |         "cvrp_run_info": [],
277 |         "ackley_run_info": {},
278 |         "rushhour_run_info": {},
279 |     }
280 |
281 |     if run_cvrp:
282 |         manifest["cvrp_run_info"] = run_final_cvrp(plan, output_dir, resume=resume)
283 |     if run_ackley:
284 |         manifest["ackley_run_info"] = run_final_ackley(plan, output_dir, resume=resume)
285 |     if run_rushhour:
286 |         manifest["rushhour_run_info"] = run_final_rushhour(plan, output_dir, resume=resume)
287 |
288 |     manifest["aggregation_info"] = aggregate_final_results(plan, output_dir)
289 |     manifest["asset_paths"] = generate_final_assets(plan, output_dir)
```

The submission audit output below is the real terminal output of the audit script on this repository — 28 checks covering files, row counts, feasibility, the report, and forbidden artifacts:

```
$ python scripts/audit_submission.py --check-results --check-pdf

PASS  file exists: README.md
PASS  file exists: requirements.txt
PASS  file exists: report/assignment3_report.md
PASS  file exists: configs/final_experiment_plan.json
PASS  file exists: data/cvrp_bks.csv
PASS  file exists: report/assignment3_report.pdf
PASS  report has no unresolved placeholders
PASS  report has no overclaim phrases
PASS  report has an AI tools section
PASS  result exists: results/final_experiments/raw/cvrp_all_instances.csv
PASS  result exists: results/final_experiments/summary/cvrp_all_summary.csv
PASS  result exists: results/final_experiments/raw/ackley_d10.csv
PASS  result exists: results/final_experiments/summary/ackley_d10_summary.csv
PASS  result exists: results/final_experiments/raw/gp_gep_comparison_runs.csv
PASS  result exists: results/final_experiments/raw/gp_gep_comparison_summary.txt
PASS  result exists: results/final_experiments/final_execution_manifest.json
PASS  cvrp_all_instances has 126 rows (126 rows)
PASS  all cvrp rows feasible
PASS  no cvrp row has errors
PASS  ackley_d10 has 21 rows (21 rows)
PASS  gp_gep_comparison has 6 rows (6 rows)
PASS  pdf size > 10 KB (452491 bytes)
PASS  forbidden absent: docs
PASS  forbidden absent: AI_USAGE.md
PASS  forbidden absent: report.md
PASS  forbidden absent: report.pdf
PASS  no ZIP files
PASS  removed instance name absent
NOTE  official .vrp files present: 6 (user-provided, not required to be tracked)

audit: PASS (0 failed check(s))
```

And the extracted report numbers, printed straight from the final CSVs
(the same numbers used in the tables above):

```
$ python scripts/extract_report_numbers.py

cvrp rows: 126, feasible: 126, with errors: 0
  best P-n16-k8: cvrp_tabu cost 451.9471 gap 0.4327%
  best E-n22-k4: cvrp_alns cost 375.2798 gap 0.0746%
  best A-n32-k5: cvrp_alns cost 787.0819 gap 0.3931%
  best A-n80-k10: cvrp_alns cost 1834.0072 gap 4.0276%
  best X-n101-k25: cvrp_aco cost 34613.1373 gap 25.4508%
  best M-n200-k17: cvrp_alns cost 1351.4311 gap 5.9946%
  mean gap baseline: 8.70%
  mean gap cvrp_aco: 7.74%
  mean gap cvrp_alns: 7.18%
  mean gap cvrp_bnb_lds: 8.70%
  mean gap cvrp_ga_island: 8.00%
  mean gap cvrp_sa: 8.63%
  mean gap cvrp_tabu: 8.12%
ackley rows: 21, errors: 0
  best ackley: ackley_alns seed 42 value 0.000000
gp/gep rows: 6
  best: gp_rushhour seed 42 eval_fitness 39956.0 solved 4
  expression diversity over all runs: 1.00
  summary txt:
    gp runs: 3
    gep runs: 3
    gp expression diversity: 1.00
    gep expression diversity: 1.00
    gep genome diversity: 1.00
    best gp expression: min((min(blocking, blocking) * (distance - 0)), ((distance * blocking) / (blocking / distance)))
    best gep expression: (5 * blocking)
    best gep genome: * 5.0 blocking min blocking log 0.5 0.0 5.0 1.0 distance 2.0 distance
    These results depend on the small puzzle set and the selected random seeds.
```

## 14. Conclusion

The implementation covers everything the assignment asks for: the six required search algorithms on both the Ackley warm-up and the six official CVRP instances, an explicit multi-stage CVRP heuristic with an honest feasibility-repair story, and separate GP and GEP frameworks for evolving Rush Hour heuristics evaluated through A*. After the vehicle-count repair work, all 126 final CVRP rows are feasible; gaps are small on the small instances (0.07–0.43%), moderate on the large ones (4–6%), and large on the capacity-tight X-n101-k25 (25.45%), which is reported as a real limitation rather than smoothed over. On Ackley, the adapted ALNS and B&B/LDS reached the optimum while untuned SA did not beat random search. GP and GEP both solved the full Rush Hour evaluation set with identical fitness, with GEP slightly faster and more compact on this small benchmark. The main open improvements are capacity-aware compound moves for tight CVRP instances, parameter tuning for SA, and a harder Rush Hour puzzle set.